Let's start in RSpec-land. Here I'll talk about the RSpec and Capybara system specs, rather than the RSpec request specs based on Rails integration tests. (Chapter 11, *Testing Rails Display Elements*, on page 217, is where you'll learn about request specs.) RSpec system specs attempt to simulate user interaction. The input to the test is typically a URL to visit, followed by simulated form fills and button clicks. The output from the test is the resulting HTML, against which the test typically searches for specific DOM selectors or text. The tests I'll talk about in this section do not interact with client-side JavaScript; in Chapter 9, *Testing JavaScript: Integration Testing*, on page 181, I'll talk about how to use feature specs to interact with a JavaScript runtime.

Setting Up Capybara

Capybara allows an automated test to simulate a user interaction with a browser. When simulating this interaction, Capybara works in conjunction with a driver, using the simple Capybara API to determine what elements to interact with and using the driver to manage the actual interaction. By default, Capybara uses a native Ruby library that doesn't manage JavaScript interactions, but it can be configured to use a headless browser such as PhantomJS, headless Chrome, or Selenium to allow JavaScript interactions to be simulated.

Capybara and RSpec

Capybara is part of the default Rails Gemfile in Rails 5.1 and later. If you're using an earlier version, add Capybara to your Gemfile's test group:

```
gem "capybara"
```

In either case, there's a useful tool called capybara-screenshot that takes virtual screenshots of any Capybara test failure.[2] Again, add capybara-screenshot in the Gemfile's test group of your Gemfile:

```
gem "capybara-screenshot"
```

Although Capybara is already available, there's a little setup needed to enable capybara-screenshot in RSpec. In a file in the spec.support directory add this:

integration/01/spec/support/system.rb

```
RSpec.configure do |config|
  config.before(:each, type: :system) do
    driven_by :rack_test
  end
end

require "capybara-screenshot/rspec"
```

2. https://github.com/mattheworiordan/capybara-screenshot